



TRAIT

How AI chatbots are evolving with distinct personalities?

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Artificial intelligence is no longer just about crunching numbers or generating automated responses. Today, AI chatbots are evolving into digital personas, responding in ways that feel eerily human-like. But what does this mean for the way we interact with technology? And more importantly, do these AI-generated personalities actually remain consistent, or are they simply a reflection of the prompts they receive?

A study titled *Do LLMs Have Distinct and Consistent Personality?*, detailed in a research paper from Yonsei University and Seoul National University, introduces TRAIT, an extensive benchmark designed to analyse the personality traits of large language models (LLMs). TRAIT employs 8,000 multi-choice questions, derived from psychometric frameworks like the Big Five Inventory (BFI) and the Short Dark Triad (SD-3), to determine whether AI

models display consistent and recognisable behavioral patterns.

The implications of these findings stretch far beyond academic curiosity. As AI chatbots become deeply embedded in customer service, personal assistants, and even mental health applications, understanding their inherent personality biases could redefine user interactions and expectations.

Breaking down TRAIT

To assess whether AI models exhibit personality traits, the TRAIT benchmark expands upon traditional psychological tests. By incorporating ATOMIC10x, a large-scale commonsense knowledge graph, researchers have tailored thousands of real-world scenarios to evaluate AI behavior in different contexts. Unlike self-assessment tests that rely on introspection, TRAIT focuses on decision-making patterns, capturing how chatbots respond to specific prompts.

The study evaluates nine leading AI models, including GPT-4, Claude Sonnet, and Mistral-7B, revealing that LLMs do exhibit distinguishable and consistent personalities. For instance, GPT-4 tends to be more agreeable than its predecessors, while alignment tuning significantly alters traits like extraversion, openness, and conscientiousness. This suggests that the datasets used for fine-tuning and reinforcement learning (RLHF) play a pivotal role in shaping an AI's personality.

However, prompting techniques – while effective in modifying AI responses – have limitations. Certain traits, such as high psychopathy or low conscientiousness, remain resistant to manipulation. This raises ethical considerations, as developers may struggle to control undesirable traits or unintended biases in AI chatbots.

AI personality in everyday life

While TRAIT provides an academic framework for understanding AI personality, its real-world impact is already visible in consumer technology. AI-driven virtual assistants like Siri, Alexa, and Google Assistant exhibit varying degrees of personality, often aligned with traits that enhance user experience. A more agreeable AI may be preferred in customer service, while a highly conscientious AI may be ideal for productivity tools. Gaming companies and social media platforms are also actively using AI personalities to drive engagement. NPCs (non-playable characters) in video games are now infused with AI-generated personalities, making interactions feel more dynamic and immersive. Similarly, AI-powered chatbots on platforms like Instagram and Snapchat are designed to maintain brand-consistent interactions, adapting their tone and language based on user input. **d**

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